Notes: Gompers, Ishii, and Metrick (QJE, 2003)

Corporate Governance and Equity Prices

Contents

1 Big picture

This paper asks whether the distribution of control rights between shareholders and managers is related to firm performance. The core idea is simple: firms differ in the number of governance provisions that restrict shareholder power and protect managers. The authors summarize these provisions in a Governance Index, denoted G, where larger values mean weaker shareholder rights and greater managerial power.

The headline result is that, during the 1990s, firms with stronger shareholder rights substantially outperformed firms with weaker shareholder rights. A portfolio that bought firms with the strongest shareholder rights, called the Democracy Portfolio, and sold firms with the weakest shareholder rights, called the Dictatorship Portfolio, earned abnormal returns of about 8.5 percent per year. The same governance index is also negatively related to Tobin's Q, profitability, and sales growth, and positively related to capital expenditures and acquisition activity.

The paper is deliberately cautious about causality. The results do not prove that governance provisions caused poor performance. Instead, the paper documents a large and robust empirical relationship, then evaluates several mechanisms: agency costs, managerial foresight, and omitted firm characteristics.

2 Research question

The main research question is:

> Do firms with stronger shareholder rights perform better than firms with weaker shareholder rights?

The paper treats corporate governance as the set of rules that determine how easily shareholders can discipline or replace managers. The key empirical challenge is that governance provisions were not randomly assigned. Firms that adopted more takeover defenses may already differ from firms that did not. Thus, the paper's evidence is strongest as a broad long-run association between governance and performance, not as a clean causal experiment.

3 Incremental contribution of the paper

The paper makes four main incremental contributions.

- **It constructs a broad firm-level governance measure.** Instead of studying a single takeover defense or a single event, the paper aggregates 24 governance provisions into one transparent index.
- **It shifts attention from short-window event studies to long-run performance.** Prior research often examined stock returns around the announcement of a new governance provision. This paper instead asks whether firms with different governance structures performed differently over the 1990s.

- **It links governance to multiple outcomes.** The paper studies stock returns, Tobin's Q, profitability, sales growth, capital expenditures, acquisitions, and insider trading.
- **It frames the causality problem explicitly.** The authors do not oversell the results as causal. They organize the interpretation around three hypotheses and use additional evidence to assess each one.

4 Data and measurement

4.1 Sample and source

The main governance data come from the Investor Responsibility Research Center (IRRC) publications *Corporate Takeover Defenses* for 1990, 1993, 1995, and 1998. The IRRC tracks large public firms, including the S&P 500 and large firms from *Fortune*, *Forbes*, and *Businessweek*. The sample covers about 1,500 firms per year and captures most of the value-weighted U.S. equity market.

The authors exclude dual-class firms because the wide variety of voting structures makes them hard to compare to ordinary single-class firms. They merge the IRRC data with CRSP for stock returns and with Compustat for accounting variables.

4.2 Governance provisions

The paper uses 24 unique governance provisions, grouped into five categories:

- **Delay:** provisions that slow hostile bidders, such as blank check preferred stock, classified boards, special meeting limits, and written consent limits.
- **Protection:** provisions that protect directors and officers, such as golden parachutes, indemnification, liability limits, and severance agreements.
- **Voting:** provisions related to shareholder voting rights, such as cumulative voting, secret ballots, supermajority rules, and unequal voting rights.
- **Other:** additional takeover-related defenses, including poison pills, fair-price provisions, antigreenmail, and directors' duties provisions.
- **State:** state takeover laws, including business combination laws, control-share acquisition laws, fair-price laws, and related statutes.

4.3 Governance Index

The Governance Index, G, is constructed by adding one point for each provision that weakens shareholder rights or increases managerial power:

$$G_i = \sum_{p=1}^{24} \mathbf{1}\{\text{provision } p \text{ weakens shareholder rights at firm } i\}.$$

A higher G means weaker shareholder rights and stronger management power. Two provisions are treated differently: Cumulative Voting and Secret Ballots are shareholder-friendly, so the index adds a point when the firm *does not* have these provisions.

The extreme portfolios are:

- **Democracy Portfolio:** firms with $G \leq 5$, interpreted as strongest shareholder rights.
- **Dictatorship Portfolio:** firms with $G \geq 14$, interpreted as weakest shareholder rights.

5 Empirical strategy

5.1 Portfolio return tests

The paper begins by comparing returns of the Democracy and Dictatorship portfolios. The benchmark performance attribution regression is the Carhart four-factor model:

$$R_t = \alpha + \beta_1 RMRF_t + \beta_2 SMB_t + \beta_3 HML_t + \beta_4 Momentum_t + \varepsilon_t.$$

Here R_t is the excess return on the portfolio being studied. In the key test, R_t is the return on a zero-investment portfolio that buys Democracy firms and shorts Dictatorship firms. The intercept α measures abnormal return relative to standard market, size, value, and momentum factors.

5.2 Firm value tests

The paper uses industry-adjusted Tobin's Q as the valuation measure:

$$Q'_{it} = a_t + bX_{it} + cW_{it} + \varepsilon_{it}.$$

Q'_{it} is firm i's Tobin's Q minus the median Q of its Fama-French industry. X_{it} is a governance variable such as G, the Democracy dummy, or subindices. W_{it} includes controls such as Delaware incorporation, log assets, log firm age, and S&P 500 membership.

5.3 Operating performance and agency-cost tests

The operating performance tests study net profit margin, return on equity, and sales growth. The agency-cost tests study capital expenditures and acquisitions. These outcomes are industry adjusted, and the authors control for book-to-market ratios to account for expected differences in investment opportunities.

5.4 Insider trading and omitted variables

The paper studies insider trading to test whether managers with many governance protections had private foresight about future poor performance. It also uses Fama-MacBeth monthly return regressions to test whether observable firm characteristics explain the return spread.

6 Main results

6.1 Return results

A dollar invested in the Dictatorship Portfolio in September 1990 grew to $3.39 by December 1999. A dollar invested in the Democracy Portfolio grew to $7.07. The annualized raw return difference is more than 9 percent per year.

After controlling for market, size, value, and momentum factors, the Democracy-minus-Dictatorship strategy earns an abnormal return of 71 basis points per month, or roughly 8.5 percent per year. The result is economically large and statistically significant.

6.2 Firm value results

Governance is negatively related to Tobin's Q. The relation becomes stronger through time. In 1990, a one-point increase in G is associated with a 2.2 percentage point lower Q. By 1999, the corresponding relation is 11.4 percentage points lower Q.

This pattern is consistent with the return evidence: during the 1990s, the market appears to increasingly penalize firms with weaker shareholder rights.

6.3 Operating performance results

Higher-G firms have lower net profit margins and lower sales growth. The effect on return on equity is weaker. The operating evidence is not as dramatic as the return evidence, but it is directionally consistent with the interpretation that weak shareholder rights are associated with poorer performance.

6.4 Agency-cost evidence

High-G firms spend more on capital expenditures and make more acquisitions. Since agency theories often predict overinvestment by entrenched managers, this evidence is consistent with the idea that weaker shareholder rights allow managers to pursue inefficient investment or acquisition activity.

6.5 No support for managerial foresight

If managers adopted governance protections because they privately foresaw poor performance, one might expect insiders at high-G firms to sell shares. The paper finds no robust relation between G and insider trading. This weakens the managerial-foresight explanation.

7 Interpretation and identification

The paper organizes interpretation around three hypotheses.

7.1 Hypothesis I: Governance causes agency costs

Under this view, weak shareholder rights make managers harder to discipline. Managers then have more room to extract private benefits, overinvest, make inefficient acquisitions, or tolerate poor performance. The evidence on lower profitability, lower sales growth, higher capital expenditures, and more acquisitions supports this hypothesis, but it does not prove it.

7.2 Hypothesis II: Governance reflects managerial foresight

Under this view, managers adopted defenses because they anticipated poor future performance. Governance protections are a symptom rather than a cause. The insider-trading evidence does not support this explanation.

7.3 Hypothesis III: Governance is correlated with omitted characteristics

Under this view, governance is proxying for some other firm trait, such as corporate culture, industry structure, organizational inertia, or some unobserved characteristic. Industry and observable controls explain part of the return spread, but not all of it. The authors conclude that unobserved characteristics remain a serious possibility.

7.4 What the paper can and cannot identify

The paper identifies large differences in realized performance across governance structures. It does not identify a clean causal treatment effect of adding or removing a governance provision. The appropriate reading is that governance is highly informative about firm performance in the 1990s, and the stakes of understanding governance causality are large.

8 Tables and figures

The paper contains tables and no standalone figures. The tables below are extracted from the original paper. Each pair of tables is placed on the same row, and the final leftover table is shown at width 0.6 of the text width.

8.1 Table I: Governance Provisions

TABLE I
GOVERNANCE PROVISIONS

	Percentage of firms with governance provisions in			
	1990	1993	1995	1998
Delay				
Blank check	76.4	80.0	85.7	87.9
Classified board	59.0	60.4	61.7	59.4
Special meeting	24.5	29.9	31.9	34.5
Written consent	24.4	29.2	32.0	33.1
Protection				
Compensation plans	44.7	65.8	72.5	62.4
Contracts	16.4	15.2	12.7	11.7
Golden parachutes	53.1	55.5	55.1	56.6
Indemnification	40.9	39.6	38.7	24.4
Liability	72.3	69.1	65.6	46.8
Severance	13.4	5.5	10.3	11.7
Voting				
Bylaws	14.4	16.1	16.0	18.1
Charter	3.2	3.4	3.1	3.0
Cumulative voting	18.5	16.5	14.9	12.2
Secret ballot	2.9	9.5	12.2	9.4
Supermajority	38.8	39.6	38.5	34.1
Unequal voting	2.4	2.0	1.9	1.9
Other				
Antigreenmail	6.1	6.9	6.4	5.6
Directors' duties	6.5	7.4	7.2	6.7
Fair price	33.5	35.2	33.6	27.8
Pension parachutes	3.9	5.2	3.9	2.2
Poison pill	53.9	57.4	56.6	55.3
Silver parachutes	4.1	4.8	3.5	2.3
State				
Antigreenmail law	17.2	17.6	17.0	14.1
Business combination law	84.3	88.5	88.9	89.9
Cash-out law	4.2	3.9	3.9	3.5
Directors' duties law	5.2	5.0	5.0	4.4
Fair price law	35.7	36.9	35.9	31.6
Control share acquisition law	29.6	29.9	29.4	26.4
Number of firms	1357	1343	1373	1708

This table presents the percentage of firms with each provision between 1990 and 1998. The data are drawn from the IRRC Corporate Takeover Defenses publications [Rosenbaum 1990, 1993, 1995, 1998] and are supplemented by data on state takeover legislation coded from Pinnell [2000]. See Appendix 1 for detailed information on each of these provisions. The sample consists of all firms in the IRRC research universe except those with dual class stock.

8.2 Table II: The Governance Index

TABLE II
THE GOVERNANCE INDEX

	1990	1993	1995	1998
Governance index				
Minimum	2	2	2	2
Mean	9.0	9.3	9.4	8.9
Median	9	9	9	9
Mode	10	9	9	10
Maximum	17	17	17	18
Standard deviation	2.9	2.8	2.8	2.8
Number of firms				
$G \leq 5$ (Democracy Portfolio)	158	139	120	215
$G = 6$	119	88	108	169
$G = 7$	158	140	127	186
$G = 8$	165	139	152	201
$G = 9$	160	183	183	197
$G = 10$	175	170	178	221
$G = 11$	149	168	166	194
$G = 12$	104	123	142	136
$G = 13$	84	100	110	106
$G = 14$ (Dictatorship portfolio)	85	93	87	83
Total	1357	1343	1373	1708
Subindex means				
Delay	1.8	2.0	2.1	2.1
Protection	2.4	2.5	2.5	2.1
Voting	2.2	2.1	2.1	2.2
Other	1.1	1.2	1.1	1.0
State	1.8	1.8	1.8	1.7

This table provides summary statistics on the distribution of G, the Governance Index, and the subindices (*Delay, Protection, Voting, Other,* and *State*) over time. G and the subindices are calculated from the provisions listed in Table I as described in Section II. Appendix 1 gives detailed information on each provision. We divide the sample into ten portfolios based on the level of G and list the number of firms in each portfolio. The Democracy Portfolio is composed of all firms where $G \leq 5$, and the Dictatorship Portfolio contains all firms where $G \geq 14$.

Read Table I:

- Table I lists the governance provisions used to build the Governance Index.
- Many provisions are very common. For example, blank check preferred stock appears in 76.4 percent of firms in 1990 and 87.9 percent in 1998.
- Business combination laws cover more than 84 percent of firms throughout the sample.
- Classified boards and poison pills are also widespread, making delay and takeover resistance central features of 1990s governance.

Economic message: The table shows that governance restrictions are not rare. They are common enough to create meaningful cross-sectional variation in shareholder rights.

Read Table II:

- The average value of G is around 9 across the sample years.
- The Democracy Portfolio contains firms with $G \leq 5$; the Dictatorship Portfolio contains firms with $G \geq 14$.
- The distribution of G is relatively stable from 1990 to 1998 despite changes in the sample.
- The subindex means show that Delay, Protection, Voting, Other, and State provisions all contribute to the overall index.

Economic message: The Governance Index compresses many governance rules into a single scalar measuring management power.

8.3 Table III: Correlations between the Subindices

TABLE III
CORRELATIONS BETWEEN THE SUBINDICES

	Delay	*Protection*	*Voting*	*Other*
Protection	0.22**			
Voting	0.33**	0.10**		
Other	0.43**	0.27**	0.19**	
State	−0.08**	−0.04	−0.07*	0.05

This table presents pairwise correlations between the subindices, *Delay, Protection, Voting, Other,* and *State* in 1990. The calculation of the subindices is described in Section II. The elements of each subindex are given in Table I and are described in detail in Appendix 1. Significance at the 5 percent and 1 percent levels is indicated by * and **, respectively.

8.4 Table IV: The Largest Firms in the Extreme Portfolios

TABLE IV
THE LARGEST FIRMS IN THE EXTREME PORTFOLIOS

1990 Democracy portfolio	State of incorporation	1990 Governance index	1998 Governance index
IBM	New York	5	6
Wal-Mart	Delaware	5	5
Du Pont de Nemours	Delaware	5	5
Pepsico	North Carolina	4	3
American International Group	Delaware	5	5
Southern Company	Delaware	5	5
Hewlett Packard	California	5	6
Berkshire Hathaway	Delaware	3	—
Commonwealth Edison	Illinois	4	6
Texas Utilities	Texas	2	4

1990 Dictatorship Portfolio	State of incorporation	1990 Governance index	1998 Governance index
GTE	New York	14	13
Waste Management	Delaware	15	13
General Re	Delaware	14	16
Limited Inc	Delaware	14	14
NCR	Maryland	14	—
K Mart	Michigan	14	10
United Telecommunications	Kansas	14	—
Time Warner	Delaware	14	13
Rorer	Pennsylvania	16	—
Woolworth	New York	14	13

This table presents the firms with the largest market capitalizations at the end of 1990 of all companies within the Democracy Portfolio ($G \leq 5$) and the Dictatorship Portfolio ($G \geq 14$). The calculation of G is described in Section II. The companies are listed in descending order of market capitalization.

Read Table III:

- Delay, Protection, Voting, and Other subindices are positively correlated with one another.
- The State subindex is weakly or negatively correlated with several firm-level subindices.
- This suggests that state law exposure is not simply the same thing as adopting firm-level takeover defenses.

Economic message: Governance provisions cluster within firms, but state laws are not a clean substitute for firm-level governance restrictions.

Read Table IV:

- The Democracy Portfolio includes large firms such as IBM, Wal-Mart, Du Pont, Pepsico, AIG, Southern Company, and Hewlett-Packard.
- The Dictatorship Portfolio includes firms such as GTE, Waste Management, General Re, Limited, NCR, KMart, Time Warner, and Woolworth.
- Both portfolios contain economically important firms; the contrast is not between tiny fringe firms and large blue-chip firms.

Economic message: The extreme portfolios are composed of large recognizable companies, making the performance differences economically meaningful.

8.5 Table V: Summary Statistics

TABLE V
SUMMARY STATISTICS

	Correlation with G	Mean, democracy portfolio	Mean, dictatorship portfolio	Difference
BM	0.02	−0.66	−0.54	−0.12
				(0.10)
SIZE	0.15**	12.86	13.46	−0.60**
				(0.21)
PRICE	0.16**	2.74	3.14	−0.40**
				(0.12)
VOLUME	0.19**	16.34	17.29	−0.95**
				(0.24)
Q	−0.04	1.77	1.47	0.30*
				(0.14)
YLD	0.03	4.20%	7.20%	−3.00%
				(4.34)
SP500	0.23**	0.15	0.49	−0.34**
				(0.06)
5-year return	−0.01	90.53%	85.41%	5.12%
				(20.74)
SGROWTH	−0.08**	62.74%	44.78%	17.96%
				(9.83)
IO	0.14**	25.89%	34.44%	−8.55%*
				(3.36)

This table gives descriptive statistics for the relationship of G with several financial and accounting measures in September 1990. The first column gives the correlations for each of these variables with the Governance Index, G. The second and third columns give means for these same variables within the Democracy Portfolio ($G \le 5$) and the Dictatorship Portfolio ($G \ge 14$) in 1990. The final column gives the difference of the two means with its standard error in parentheses. The calculation of G is described in Section II, and definitions of each variable are given in Appendix 2. Significance at the 5 percent and 1 percent levels is indicated by * and **, respectively.

8.6 Table VI: Performance-Attribution Regressions for Decile Portfolios

TABLE VI
PERFORMANCE-ATTRIBUTION REGRESSIONS FOR DECILE PORTFOLIOS

	α	*RMRF*	*SMB*	*HML*	*Momentum*
Democracy-Dictatorship	**0.71****	**−0.04**	**−0.22***	**−0.55***	**−0.01**
	(0.26)	**(0.07)**	**(0.09)**	**(0.10)**	**(0.07)**
$G \le 5$ (Democracy)	0.29*	0.98**	−0.24**	−0.21**	−0.05
	(0.13)	(0.04)	(0.05)	(0.05)	(0.03)
$G = 6$	0.22	0.99**	−0.18**	0.05	−0.08
	(0.18)	(0.05)	(0.06)	(0.07)	(0.04)
$G = 7$	0.24	1.05**	−0.10	−0.14	0.15**
	(0.19)	(0.05)	(0.07)	(0.08)	(0.05)
$G = 8$	0.08	1.02**	−0.04	−0.08	0.01
	(0.14)	(0.04)	(0.05)	(0.06)	(0.04)
$G = 9$	−0.02	0.97**	−0.20**	0.14**	−0.01
	(0.12)	(0.03)	(0.04)	(0.05)	(0.03)
$G = 10$	0.03	0.95**	−0.17**	−0.00	−0.08**
	(0.11)	(0.03)	(0.04)	(0.04)	(0.03)
$G = 11$	0.18	0.99**	−0.14*	−0.06	−0.01
	(0.16)	(0.05)	(0.05)	(0.06)	(0.04)
$G = 12$	−0.25	1.00**	−0.11*	0.16**	0.02
	(0.14)	(0.04)	(0.05)	(0.06)	(0.04)
$G = 13$	−0.01	1.03**	−0.21**	0.14*	−0.08*
	(0.14)	(0.04)	(0.05)	(0.06)	(0.04)
$G \ge 14$ (Dictatorship)	−0.42*	1.03**	−0.02	0.34**	−0.05
	(0.19)	(0.05)	(0.06)	(0.07)	(0.05)

We estimate four-factor regressions (equation (1) from the text) of value-weighted monthly returns for portfolios of firms sorted by G. The calculation of G is described in Section II. The first row contains the results when we use the portfolio that buys the Democracy Portfolio ($G \le 5$) and sells short the Dictatorship Portfolio ($G \ge 14$). The portfolios are reset in September 1990, July 1993, July 1995, and February 1998, which are the months after new data on G became available. The explanatory variables are *RMRF*, *SMB*, *HML*, and *Momentum*. These variables are the returns to zero-investment portfolios designed to capture market, size, book-to-market, and momentum effects, respectively. (Consult Fama and French [1993] and Carhart [1997] on the construction of these factors.) The sample period is from September 1990 through December 1999. Standard errors are reported in parentheses and significance at the 5 percent and 1 percent levels is indicated by * and **, respectively.

Read Table V:

- High-G firms are more likely to be in the S&P 500 and tend to be larger, have higher share prices, and higher trading volume.
- Democracy firms have higher average Tobin's Q than Dictatorship firms in 1990.
- Dictatorship firms have lower past five-year sales growth than Democracy firms.

Economic message: Governance is correlated with firm characteristics, so raw return differences must be interpreted carefully and controlled for.

Read Table VI:

- The Democracy-minus-Dictatorship portfolio earns a four-factor alpha of 0.71 percent per month.
- The Democracy Portfolio alone has positive alpha, while the Dictatorship Portfolio alone has negative alpha.
- Alpha generally declines as G increases.

Economic message: The main return result is not just raw performance. It survives standard controls for market, size, value, and momentum exposure.

8.7 Table VII: Performance-Attribution Regressions under Alternative Portfolio Constructions

TABLE VII
PERFORMANCE-ATTRIBUTION REGRESSIONS UNDER ALTERNATIVE PORTFOLIO CONSTRUCTIONS

	α, Value-weighted	α, Equal-weighted
(1) Democracy-Dictatorship	0.71**	0.45*
	(0.26)	(0.22)
(2) Industry-adjusted	0.47*	0.30
	(0.22)	(0.19)
(3) Big portfolios	0.47*	0.39*
	(0.21)	(0.19)
(4) Small portfolios	0.78*	0.45
	(0.33)	(0.25)
(5) 1990 portfolio	0.53*	0.33
	(0.24)	(0.22)
(6) Delaware portfolio	0.63	0.42
	(0.34)	(0.26)
(7) Early half	0.45	0.58*
	(0.23)	(0.28)
(8) Late half	0.75	0.04
	(0.40)	(0.27)

This table presents the alphas from four-factor regressions for variations on the Democracy ($G \leq 5$) minus Dictatorship ($G \geq 14$) Portfolio. The calculation of G is described in Section II. The portfolios are reset in September 1990, July 1993, July 1995, and February 1998, which are the months after new data on G became available. The sample period is September 1990 to December 1999. The first row uses the unadjusted difference between the monthly returns to the Democracy and Dictatorship Portfolios. The second row contains the results using industry-adjusted returns, with industry adjustments done relative to the 48 industries of Fama and French [1997]. The third and fourth rows use alternative definitions of the Democracy and Dictatorship Portfolios. In the third row, firms are sorted on G and the two portfolios contain the smallest set of firms with extreme values of G such that each has at least 10 percent of the sample. This implies cutoff values of G for the Democracy Portfolio of 5, 5, 6, and 5 for September 1990, July 1993, July 1995, and February 1998, respectively. The cutoffs for the Dictatorship Portfolio are always 13. In the fourth row, the two portfolios contain the largest set of firms such that each has no more than 10 percent of the sample. The cutoff values of G for the Democracy Portfolio are 4, 4, 5, and 4 for September 1990, July 1993, July 1995, and February 1998, respectively, and they are always 14 for the Dictatorship Portfolio. In the fifth row, portfolio returns are calculated maintaining the 1990 portfolios for the entire sample period. As long as they are listed in CRSP, we neither delete nor add firms to these portfolios regardless of subsequent changes in G or changes in the IRRC sample in later editions. The sixth row shows the results of restricting the sample to firms incorporated in Delaware. In the seventh and eighth rows, the sample period is divided in half at April 30, 1995, and separate regressions are estimated for the first half and second half of the period (56 months each). The explanatory variables are *RMRF*, *SMB*, *HML*, *Momentum*, and a constant. These variables are the returns to zero-investment portfolios designed to capture market, size, book-to-market, and momentum effects, respectively. (Consult Fama and French [1993] and Carhart [1997] on the construction of these factors.) All coefficients except for the alpha are omitted in this table. Standard errors are reported in parentheses, and significance at the 5 percent and 1 percent levels is indicated by * and **, respectively.

8.8 Table VIII: Q Regressions

TABLE VIII
Q REGRESSIONS

	(1) G	(2) *Democracy Portfolio*	(3) *Delay*	(4) *Protection*	(5) *Voting*	(6) *Other*	(7) *State*
1990	−0.022**	0.186	−0.015	−0.035	0.015	−0.031	−0.004
	(0.008)	(0.127)	(0.022)	(0.018)	(0.030)	(0.026)	(0.020)
1991	−0.040**	0.302*	−0.033	−0.048	−0.012	−0.059	0.003
	(0.012)	(0.143)	(0.034)	(0.028)	(0.047)	(0.040)	(0.031)
1992	−0.036**	0.340*	−0.041	−0.039	0.021	−0.054	−0.011
	(0.010)	(0.151)	(0.027)	(0.023)	(0.038)	(0.032)	(0.025)
1993	−0.042**	0.485*	−0.023	−0.055*	0.009	−0.060	−0.062*
	(0.011)	(0.204)	(0.029)	(0.026)	(0.038)	(0.035)	(0.027)
1994	−0.031**	0.335*	−0.032	−0.012	−0.032	−0.029	−0.047*
	(0.009)	(0.161)	(0.023)	(0.020)	(0.031)	(0.028)	(0.022)
1995	−0.039**	0.435*	−0.046	−0.062*	−0.086*	0.023	−0.022
	(0.011)	(0.217)	(0.030)	(0.027)	(0.041)	(0.036)	(0.028)
1996	−0.025*	0.299	−0.029	−0.030	−0.078	0.018	−0.024
	(0.011)	(0.195)	(0.031)	(0.028)	(0.041)	(0.037)	(0.028)
1997	−0.016	0.210	−0.017	−0.007	−0.055	−0.001	−0.017
	(0.013)	(0.196)	(0.035)	(0.032)	(0.047)	(0.042)	(0.032)
1998	−0.065**	0.203	−0.023	−0.096*	−0.132	−0.058	0.012
	(0.020)	(0.404)	(0.052)	(0.049)	(0.070)	(0.066)	(0.052)
1999	−0.114**	0.564	−0.067	−0.171*	−0.294**	−0.006	−0.033
	(0.027)	(0.602)	(0.071)	(0.067)	(0.098)	(0.090)	(0.073)
Mean	−0.043**	0.336**	−0.033**	−0.056**	−0.065	−0.025*	−0.020*
	(0.009)	(0.040)	(0.005)	(0.015)	(0.030)	(0.010)	(0.007)

The first column of this table presents the coefficients on G, the Governance Index, from regressions of industry-adjusted Tobin's Q on G and control variables. The second column restricts the sample to firms in the Democracy ($G \leq 5$) and Dictatorship ($G \geq 14$) Portfolios and includes as regressors a dummy variable for the Democracy Portfolio and the controls. The third through seventh columns show the coefficients on each subindex from regressions where the explanatory variables are the subindices *Delay*, *Protection*, *Voting*, *Other*, and *State*, and the controls. We include as controls a dummy variable for incorporation in Delaware, the log of assets in the current fiscal year, the log of firm age measured in months as of December of each year, and a dummy variable for inclusion in the S&P 500 as of the end of the previous year. The coefficients on the controls and the constant are omitted from the table. The calculation of G and the subindices is described in Section II. Q is the ratio of the market value of assets to the book value of assets: the market value is calculated as the sum of the book value of assets and the market value of common stock less the book value of common stock and deferred taxes. The market value of equity is measured at the end of the current calendar year, and the accounting variables are measured in the current fiscal year. Industry adjustments are made by subtracting the industry median, where medians are calculated by matching the four-digit SIC codes from December of each year to the 48 industries designated by Fama and French [1997]. The coefficients and standard errors from each annual cross-sectional regression are reported in each row, and the time-series averages and time-series standard errors are given in the last row. * and ** indicate significance at the 5 percent and 1 percent levels, respectively.

cients and standard errors for a different year of the sample; the last row gives the average coefficient and time-series standard error of these coefficients. The coefficients on G are negative in

Read Table VII:

- The Democracy-minus-Dictatorship alpha remains positive under value-weighted and equal-weighted constructions.
- Industry adjustment reduces but does not eliminate the return spread.
- Keeping the 1990 portfolios fixed still produces a positive alpha.
- The Delaware-only test also yields a positive alpha, though with weaker statistical precision.

Economic message: The abnormal return finding is not driven solely by portfolio cutoffs, industry composition, or later changes in governance data.

Read Table VIII:

- The coefficient on G in Q regressions is negative in every year.
- The magnitude grows over the decade, reaching −0.114 in 1999.
- The Democracy dummy is generally positive, implying higher Q for strong-shareholder-rights firms.

- Subindex coefficients are usually negative, but multicollinearity makes it hard to identify which type of provision matters most.

Economic message: Firm valuation increasingly reflects governance differences over the 1990s.

8.9 Table IX: Operating Performance

TABLE IX
OPERATING PERFORMANCE

	(1)	(2)	(3)	(4)	(5)	(6)
	Net profit margin		Return on equity		Sales growth	
	G	*Democracy Portfolio*	G	*Democracy Portfolio*	G	*Democracy Portfolio*
1991	−0.70	10.61	−1.19*	13.54	−2.30	−3.52
	(0.39)	(7.12)	(0.60)	(11.30)	(1.38)	(17.83)
1992	−0.52	9.45	0.42	2.54	−1.43	0.10
	(0.58)	(10.43)	(0.61)	(9.21)	(1.06)	(11.52)
1993	−0.76	7.77	−0.34	2.51	−3.35**	18.55
	(0.48)	(9.98)	(0.79)	(10.98)	(1.17)	(17.71)
1994	−0.83	10.94	−1.07	2.69	−2.71*	12.58
	(0.48)	(6.59)	(0.61)	(10.36)	(1.10)	(22.81)
1995	−0.72	7.56	−1.39	14.77	−0.89	7.91
	(0.67)	(8.30)	(0.75)	(9.88)	(1.70)	(19.67)
1996	−0.43	−2.17	0.90	−2.30	−2.44	14.84
	(0.40)	(7.22)	(0.65)	(12.09)	(1.39)	(19.36)
1997	0.21	−9.61	0.66	−17.54	0.01	−4.28
	(0.55)	(9.99)	(0.81)	(9.83)	(1.64)	(26.61)
1998	−0.73	−3.99	−1.28	13.62	−1.45	−15.65
	(0.63)	(7.15)	(1.01)	(15.10)	(1.50)	(23.36)
1999	−1.27*	4.59	0.93	−15.53	−0.52	15.38
	(0.58)	(11.58)	(0.85)	(10.38)	(1.92)	(26.10)
Mean	−0.64**	3.91	−0.26	1.59	−1.68**	5.10
	(0.13)	(2.46)	(0.33)	(3.98)	(0.37)	(3.84)

The first, third, and fifth columns of this table give the results of annual median (least absolute deviation) regressions for net profit margin, return on equity, and sales growth on the Governance Index, G, measured in the previous year, and the book-to-market ratio, BM. The second, fourth, and sixth columns restrict the sample to firms in the Democracy ($G \leq 5$) and Dictatorship ($G \geq 14$) portfolios and include as regressors a dummy variable for the Democracy Portfolio and BM. The coefficients on BM and the constant are omitted from the table. The calculation of G is described in Section II. Net profit margin is the ratio of income before extraordinary items available for common equity to sales; return on equity is the ratio of income before extraordinary items available for common equity to the sum of the book value of common equity and deferred taxes; BM is the log of the ratio of book value (the sum of book common equity and deferred taxes) in the previous fiscal year to size at the close of the previous calendar year. Each dependent variable is net of the industry median, which is calculated by matching the four-digit SIC codes of all firms in the CRSP-Compustat merged database in December of each year to the 48 industries designated by Fama and French [1997]. The coefficients and standard errors from each annual cross-sectional regression are reported in each row, and the time-series averages and time-series standard errors are given in the last row. Significance at the 5 percent and 1 percent levels is indicated by * and **, respectively. All coefficients and standard errors are multiplied by 1000.

8.10 Table X: Capital Expenditure

TABLE X
CAPITAL EXPENDITURE

	(1)	(2)	(3)	(4)
	CAPEX/Assets		CAPEX/Sales	
	G	*Democracy Portfolio*	G	*Democracy Portfolio*
1991	1.32**	−13.02**	0.70*	−9.28
	(0.27)	(4.28)	(0.32)	(4.96)
1992	0.42	−7.03	0.54	−7.23
	(0.35)	(4.86)	(0.35)	(6.01)
1993	0.81*	−6.06	0.09	−1.68
	(0.37)	(4.48)	(0.34)	(4.98)
1994	0.51	−7.84	−0.07	−4.82
	(0.32)	(5.21)	(0.37)	(4.76)
1995	0.35	−3.40	0.32	−9.80
	(0.39)	(6.83)	(0.39)	(5.90)
1996	0.75	−6.90	0.31	−3.26
	(0.39)	(5.55)	(0.33)	(6.36)
1997	0.74*	−4.23	0.70	−8.05
	(0.34)	(3.50)	(0.40)	(5.71)
1998	0.80*	−10.57	0.37	−6.43
	(0.37)	(6.75)	(0.35)	(5.63)
1999	−0.15	3.12	−0.32	3.49
	(0.39)	(4.20)	(0.38)	(5.52)
Mean	0.62**	−6.21**	0.30*	−5.23**
	(0.13)	(1.53)	(0.11)	(1.41)

The first and third columns of this table present the results of annual median (least absolute deviation) regressions of CAPEX/Assets and CAPEX/Sales on the Governance Index, G, measured in the previous year, and BM. The second and fourth columns restrict the sample to firms in the Democracy ($G \leq 5$) and Dictatorship ($G \geq 14$) portfolios and include as regressors a dummy variable for the Democracy portfolio and BM. The coefficients of BM and the constant are omitted from the table. The calculation of G is described in Section II. CAPEX is capital expenditures, and BM is the log of the ratio of book value (the sum of book common equity and deferred taxes) in the previous fiscal year to size at the close of the previous calendar year. Both dependent variables are net of the industry median, which is calculated by matching the four-digit SIC codes of all firms in the CRSP-Compustat merged database in December of each year to the 48 industries designated by Fama and French [1997]. The coefficients and standard errors from each annual cross-sectional regression are reported in each row, and the time-series averages and time-series standard errors are given in the last row. Significance at the 5 percent and 1 percent levels is indicated by * and **, respectively. All coefficients and standard errors are multiplied by 1000.

other things equal, high-G firms have higher CAPEX than do low-G firms.

Another outlet for capital expenditure is for firms to acquire

Read Table IX:

- Higher G is negatively related to net profit margin on average.
- Higher G is also negatively related to sales growth.
- The return-on-equity results are weaker and less consistent.

Economic message: Firms with weaker shareholder rights also appear to have weaker operating performance, not just lower stock returns.

Read Table X:

- Higher G is positively related to capital expenditures scaled by assets and by sales.
- Democracy firms have lower capital expenditure than Dictatorship firms in the extreme-portfolio specification.
- These patterns are consistent with overinvestment by more entrenched managers.

Economic message: The table provides agency-cost evidence: weak governance is associated with higher investment spending.

8.11 Table XI: Acquisitions

TABLE XI
ACQUISITIONS

	(1)	(2)	(3)	(4)
	Acquisition count (Poisson regressions)		*Acquisition ratio (Tobit regressions)*	
	G	*Democracy Portfolio*	*G*	*Democracy Portfolio*
1991	1.58	−50.81	0.51	0.14
	(1.46)	(26.12)	(0.47)	(5.03)
1992	1.64	−31.39	0.10	7.91
	(1.44)	(24.61)	(0.50)	(6.42)
1993	1.75	−47.67	0.70	−6.31
	(1.42)	24.51	(0.56)	(6.85)
1994	4.09**	−13.10	0.75	1.82
	(1.27)	(21.02)	(0.48)	(4.14)
1995	2.57*	−60.92**	0.41	−2.95
	(1.15)	(17.85)	(0.44)	(4.42)
1996	2.69*	−66.06**	1.33*	−24.22**
	(1.14)	(20.48)	(0.60)	(9.41)
1997	2.34*	−63.81**	0.99*	−9.24
	(1.12)	(19.03)	(0.51)	(6.78)
1998	2.42*	−52.03**	1.47	−11.11
	(1.09)	(17.67)	(0.76)	(8.51)
1999	0.52	−47.64**	0.84	−20.87*
	(1.01)	(17.27)	(0.74)	(9.68)
Mean	2.18**	−48.16**	0.79**	−7.21
	(0.33)	(5.60)	(0.14)	(3.49)

The first column of this table presents annual Tobit regressions of the *Acquisition ratio* on the Governance Index, G, measured in the previous year, *SIZE*, *BM*, and industry dummy variables. The third column presents annual Poisson regressions of *Acquisition count* on the same explanatory variables. In the second and fourth columns, we restrict the sample to firms in the Democracy ($G \leq 5$) and Dictatorship ($G \geq 14$) Portfolios, and we include as a regressor a dummy variable that equals 1 when the firm is in the Democracy Portfolio and 0 otherwise. The coefficients on *SIZE*, *BM*, and the industry dummy variables are omitted from the table. The calculation of G is described in Section II. *Acquisition ratio* is defined as the sum of the value of all corporate acquisitions during a calendar year scaled by the average of market value at the beginning and end of the year. *Acquisition count* is defined as the number of acquisitions during a calendar year. The data on acquisitions are from the SDC database. *SIZE* is the log of market capitalization at the end of the previous calendar year in millions of dollars, and *BM* is the log of the ratio of book value (the sum of book common equity and deferred taxes) in the previous fiscal year to size at the close of the previous calendar year. Industry dummy variables are created by matching the four-digit SIC codes of all firms in the CRSP-Compustat merged database in December of each year to the 48 industries designated by Fama and French [1997]. The coefficients and standard errors from each annual cross-sectional regression are reported in each row, and the time-series averages and time-series standard errors are given in the last row. Significance at the 5 percent and 1 percent levels is indicated by * and **, respectively. All coefficients and standard errors are multiplied by 100.

8.12 Table XII: Insider Trading

TABLE XII
INSIDER TRADING

	(1)	(2)	(3)	(4)
	OLS		Ordered logit	
	G	*Democracy Portfolio*	*G*	*Democracy Portfolio*
1991	0.07*	−0.14	−8.85	−345.18
	(0.04)	(0.53)	(21.34)	(295.15)
1992	0.10	−1.47	−66.92**	499.93
	(0.07)	(1.50)	(21.70)	(310.53)
1993	0.10	−0.23	−32.40	797.17*
	(0.07)	0.51	(21.41)	(326.87)
1994	0.07	−0.61	−28.09	323.07
	(0.04)	(1.23)	(20.58)	(290.11)
1995	0.04	−0.17	−4.66	−153.33
	(0.02)	(0.20)	(22.00)	(308.90)
1996	0.15	−0.62	12.01	−93.95
	(0.14)	(1.05)	(21.67)	(321.18)
1997	−0.01	0.89	−46.08	781.42*
	(0.10)	(0.66)	(24.33)	(369.78)
1998	−0.12	2.41	−1.88	146.49
	(0.20)	(3.17)	(24.31)	(342.22)
1999	0.36	−1.36	4.41	−117.36
	(0.48)	(2.91)	(21.09)	(323.85)
Mean	0.09	−0.15	−19.16	204.25
	(0.04)	(0.40)	(8.66)	(140.02)

The first and third columns of this table present annual OLS and ordered logit regressions of *Net insider purchases* on G measured in the previous year, *SIZE*, *BM*, and a constant. In the second and fourth columns, we restrict the sample to firms in the Democracy ($G \leq 5$) and Dictatorship ($G \geq 14$) Portfolios and we include as a regressor a dummy variable that equals 1 when the firm is in the Democracy Portfolio and 0 otherwise. The coefficients on *SIZE*, *BM*, and the constant are omitted from the table. The calculation of G is described in Section II. *Net insider purchases* is the sum of split-adjusted open market purchases less split-adjusted open market sales during a year scaled by shares outstanding at the end of the previous calendar year. The ordered logit regressions use a dependent variable that equals 1 if *Net insider purchases* is positive, 0 if it is zero, and −1 if it is negative. The data on insider sales are from the Thomson database. *SIZE* is the log of market capitalization in millions of dollars measured at the end of the previous calendar year, and *BM* is the log of the ratio of book value (the sum of book common equity and deferred taxes) in the previous fiscal year to size at the close of the previous calendar year. The coefficients and standard errors from each annual cross-sectional regression are reported in each row, and the time-series averages and time-series standard errors are given in the last row. Significance at the 5 percent and 1 percent levels is indicated by * and **, respectively. All coefficients and standard errors are multiplied by 1000.

Read Table XI:

- Higher G predicts a greater acquisition count.
- Higher G also predicts a higher acquisition ratio.
- Democracy firms tend to make fewer acquisitions than Dictatorship firms.

Economic message: Weak-shareholder-rights firms are more acquisition-active, consistent with a possible agency channel through inefficient expansion or attempted prevention of empire collapse.

Read Table XII:

- The relation between governance and insider trading is not robust.
- Average coefficients are not statistically significant.
- There is no strong evidence that insiders at weak-governance firms sold stock before poor performance.

Economic message: The insider trading evidence weakens the hypothesis that governance provisions merely reflect managers' private forecasts of future poor performance.

8.13 Table XIII: Fama-MacBeth Return Regressions

TABLE XIII
FAMA-MACBETH RETURN REGRESSIONS

	(1) Raw	(2) Industry-adjusted	(3) Raw	(4) Industry-adjusted	(5) Raw	(6) Industry-adjusted
G	−0.04	−0.02				
	(0.04)	(0.03)				
Democracy Portfolio			0.76*	0.63*		
			(0.32)	(0.26)		
Delay					−0.03	0.02
					(0.10)	(0.07)
Protection					−0.07	−0.01
					(0.08)	(0.06)
Voting					−0.08	−0.08
					(0.13)	(0.10)
Other					0.01	−0.04
					(0.08)	(0.07)
State					0.02	−0.04
					(0.08)	(0.06)
NASDUM	−0.83	−0.42	−8.23	−10.36	−2.60	−0.29
	(6.94)	(5.26)	(6.45)	(5.94)	(6.39)	(4.98)
SP500	−0.19	−0.20	−0.42	−0.21	−0.19	−0.24
	(0.49)	(0.42)	(0.49)	(0.41)	(0.45)	(0.40)
BM	0.04	0.14	0.06	0.11	0.06	0.15
	(0.19)	(0.12)	(0.38)	(0.29)	(0.20)	(0.11)
SIZE	0.17	0.22	0.47	0.02	0.19	0.24
	(0.27)	(0.16)	(0.38)	(0.32)	(0.27)	(0.17)
PRICE	0.26	0.20	0.28	0.44	0.20	0.16
	(0.26)	(0.20)	(0.31)	(0.31)	(0.28)	(0.22)
IO	0.61	0.10	0.78	−0.16	0.59	0.14
	(0.47)	(0.33)	(0.67)	(0.60)	(0.44)	(0.33)
NYDVOL	−0.11	−0.21	−0.49	−0.03	−0.13	−0.21
	(0.29)	(0.18)	(0.36)	(0.31)	(0.28)	(0.18)
NADVOL	0.01	−0.13	−0.09	0.48	0.06	−0.15
	(0.43)	(0.29)	(0.41)	(0.39)	(0.43)	(0.29)
YLD	10.85	10.94	15.74	9.23	6.21	8.76
	(10.54)	(7.25)	(14.62)	(11.56)	(11.63)	(7.70)
RET2–3	−0.48	−0.93	−2.04	−1.82	−0.57	−1.03
	(1.40)	(1.04)	(2.33)	(1.73)	(1.43)	(1.07)
RET4–6	−0.68	−0.48	−2.21	−1.12	−0.58	−0.55
	(1.33)	(0.92)	(1.89)	(1.36)	(1.33)	(0.93)
RET7–12	2.42*	0.89	0.12	−1.67	2.69**	1.06
	(1.00)	(0.65)	(1.35)	(1.03)	(0.99)	(0.65)
SGROWTH	−0.00	0.03	0.75	0.27	−0.01	0.02
	(0.26)	(0.18)	(0.47)	(0.40)	(0.25)	(0.18)
Constant	−0.53	−0.18	1.17	−1.86	0.03	−0.16
	(2.55)	(1.71)	(3.43)	(2.99)	(2.39)	(1.69)

This table presents the average coefficients and time-series standard errors for 112 cross-sectional regressions for each month from September 1990 to December 1999. The dependent variable is the stock return for month t. The results are presented using both raw and industry-adjusted returns, with industry adjustments done using the 48 industries of Fama and French [1997]. The first and second columns include all firms with data for all right-hand side variables and use G, the Governance index, as an independent variable. In the third and fourth columns, the sample is restricted to firms in either the Democracy ($G \leq 5$) or Dictatorship ($G \geq 14$) Portfolios, and we use the independent variable, *Democracy Portfolio*, a dummy variable that equals 1 when the firm is in the Democracy Portfolio and 0 otherwise. In the fifth and sixth columns, we again include all firms with data for each explanatory variable and use the subindices, *Delay*,

Read Table XIII:

- In firm-level monthly cross-sectional regressions, the Democracy dummy remains positive and significant.
- Industry-adjusted returns reduce the effect but do not eliminate it.
- The overall G coefficient is negative but not statistically significant after many controls.
- Subindex coefficients are imprecise, so the paper cannot isolate a single category of provisions as the unique driver.

Economic message: Observable controls explain some but not all of the return gap. Governance remains economically important, but causal interpretation remains limited.

9 Short summary

Gompers, Ishii, and Metrick construct a Governance Index using 24 provisions that restrict or strengthen shareholder rights. During the 1990s, firms with strong shareholder rights outperform firms with weak shareholder rights by about 8.5 percent per year after risk-factor controls. Strong-rights firms also have higher Tobin's Q, higher operating performance, lower capital expenditures, and fewer acquisitions. The evidence is consistent with an agency-cost interpretation, but the authors emphasize that the design is not randomized and cannot fully rule out omitted variables.

10 Conclusion

The paper's central conclusion is that governance structure is strongly associated with equity prices and corporate performance. The Governance Index captures a meaningful dimension of managerial power: firms with more restrictions on shareholder rights performed substantially worse during the 1990s.

The most natural interpretation is that weaker shareholder rights increase agency costs by insulating managers from discipline. The supporting evidence comes from poorer operating performance, higher capital expenditures, and more acquisition activity at high-G firms. However, the paper does not provide a clean causal design. Governance provisions may proxy for deeper firm characteristics or corporate culture.

Takeaway: The paper establishes that corporate governance is empirically important and economically large. It also makes clear why identifying the causal effect of governance rules is difficult: governance is chosen by firms, not randomly assigned.